

E1

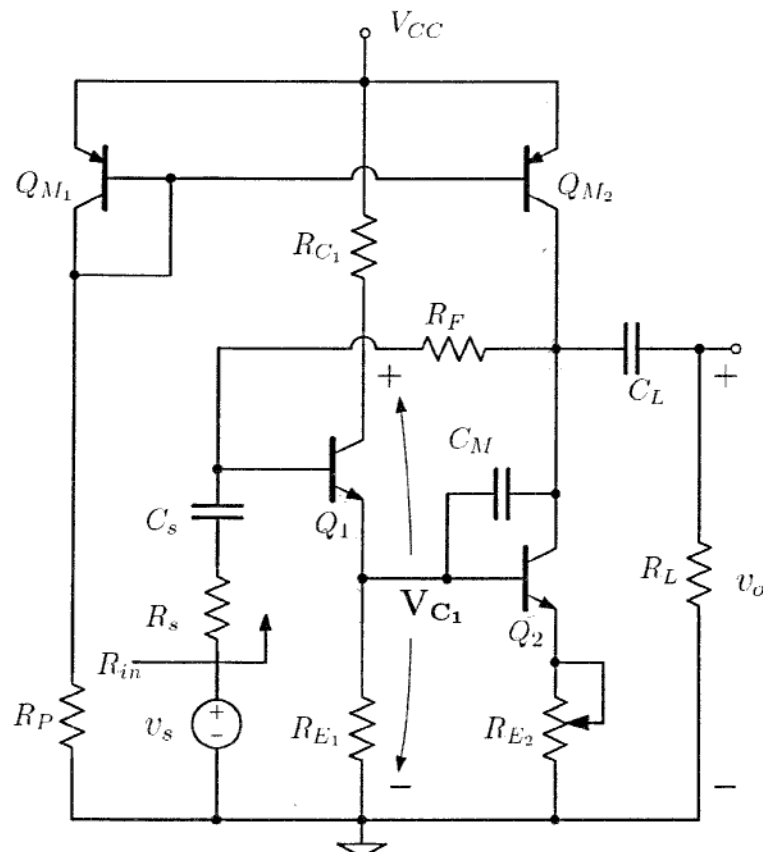


Fig. 1 – A case of two-stage amplifier with feedback.

Consider the circuit in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{CC} = 10\text{ V}$; $R_P = 4.7\text{ k}\Omega$; $R_{C1} = 1\text{ k}\Omega$; $R_{E1} = 1\text{ k}\Omega$; $R_F = 200\text{ k}\Omega$; $R_s = 10\text{ k}\Omega$; $R_L = 18\text{ k}\Omega$.
 $C_s = 1\text{ }\mu\text{F}$; $C_L = 1\text{ }\mu\text{F}$; $C_M = 50\text{ pF}$.

$Q_{1,2}$ (nnp BJT): $V_{BE1,2} = 0.65\text{ V}$; $\beta_{F1,2} = \beta_{01,2} = 200$; $r_{01,2} = +\infty$;

$Q_{M1,2}$ (pnp BJT): $V_{BEM1,2} = -0.7\text{ V}$; $\beta_{FM1,2} = \beta_{0M1,2} = 150$; $r_{0M1,2} = +\infty$;

$V_T = 26\text{ mV}$.

Considering all capacitors to be equivalent to *open circuits*, determine:

1. the value of R_{E2} that makes the bias level of the collector of Q_1 , V_{C1} , equal to 8 V ;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), C_s and C_L are equivalent to *short circuits*, while C_M is equivalent to an *open circuit*, determine:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance indicated in the figure.

Assuming only C_M is effective (while C_s, C_L are shorted) and neglecting all C_{π} 's and C_{μ} 's, determine:

5. the high frequency bandwidth limit f_H of the amplifier.

E2

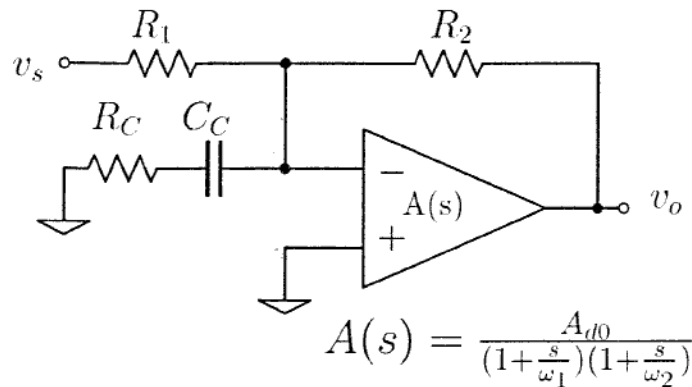


Fig. 2 – Non ideal, OpAmp based, inverting amplifier.

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 56 \text{ k}\Omega$; $R_2 = 120 \text{ k}\Omega$; $R_C = 470 \text{ }\Omega$; $C_C = 100 \text{ nF}$;

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$; $\omega_1 = 1 \text{ krad/s}$; $\omega_2 = 500 \text{ krad/s}$.

Considering, initially, C_C to be open circuited, determine:

1. the expression of the *ideal* amplifier voltage gain $A_v(s) = v_o/v_s$;
2. the equivalent *block diagram* of the amplifier, according to *feedback theory*;
3. the amplifier *bandwidth* and *phase margin*.

Then, applying the noise gain compensation network R_C - C_C , find:

4. the *bandwidth* and the *phase margin* of the compensated amplifier.